

Full Stack Development with SQL Server, Tableau & Flask

Project Documentation Format

1. Introduction

- Project Title: Plugging Into the Future: An Exploration of Electricity Consumption Patterns
- Team Members: [List team members and their roles]

2. Project Overview

Purpose: To clean, structure, and visualize state-wise electricity consumption data so trends and comparisons are easy to explore.

Features: SQL Server data pipeline, aggregation queries, interactive Tableau dashboards and story, and a Flask web app that presents the insights.

3. Architecture

- Frontend: HTML/CSS templates served by Flask, embedding the Tableau dashboards.
- Backend: Flask (Python) handling routing and page serving.
- Database: Microsoft SQL Server — schema holds the cleaned, aggregated consumption tables queried by Tableau.

4. Setup Instructions

- Prerequisites: Python 3.x, Flask, Microsoft SQL Server (with the project database restored), Tableau (Desktop or Public account).
- Installation: Clone the repository, run ``pip install -r requirements.txt``, configure the SQL Server connection string, then run the app.

5. Folder Structure

- `app/` — Flask application (routes, templates, static files).
- `sql/` — SQL Server scripts for schema creation, cleaning, and aggregation queries.
- `tableau/` — Tableau workbook(s) and published dashboard/story links.

6. Running the Application

- Backend: ``flask run`` (or ``python app.py``) from the project root.

7. API Documentation

The Flask app primarily serves dashboard/report pages rather than a JSON API; document any additional routes here as they're added.

8. Authentication

Not implemented in the current scope — the application serves read-only public dashboards.

9. User Interface

See the screenshots in the Doc and Demo / Results sections of the final report.

10. Testing

See the Performance Testing and User Acceptance Testing documents in this project's Project Development Phase folder.

11. Screenshots or Demo

See Video_Demo.mp4 in the Project Demonstration folder.

12. Known Issues

[Document any known bugs or issues here.]

13. Future Enhancements

Potential additions include automated data refresh, more granular (district-level) data, and public API access to the aggregated dataset.